

NG YI XUAN

Software Engineer

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Triang, Pahang (relocating to Kuala Lumpur)

SUMMARY

Final-year Computer Science (Network Engineering) student at Universiti Malaysia Sabah, CGPA 3.51, with a six-month full-stack internship. I build web apps with React and Laravel, including real-time features over WebSockets and AI features using LLM APIs. Available to start full-time from December 2026.

EDUCATION

**Bachelor of Computer Science with Honours
(Network Engineering)**
Universiti Malaysia Sabah
Graduating Nov 2026

CGPA
3.51 / 4.00

- Software:** Web Technology, Database, Software Engineering, System Analysis & Design, Object-Oriented Programming, Data Structures & Algorithms
- Systems & data:** Cloud Computing (virtualization, VPC), Big Data Analytics (SQL/NoSQL, Spark, R), Cybersecurity, Network Programming

EXPERIENCE

Software Engineering Intern
Go Digital & Automation (placed at EAN Label Industry) 03/2026 to 08/2026

- Stock in the warehouse system had not matched the shelves for years. Traced the mismatches with SQL and cleaned the historical data until the system reported stock accurately.
- Rebuilt old internal screens as simpler dashboards in React and Nuxt.js, and linked the company ERP to the warehouse system at the database and backend level so their data stopped drifting apart.
- Proposed and built a reporting dashboard that shows Finance stock quantity, value, movement and ageing; the team adopted it.
- Worked in a team of five. Used GitHub Copilot and Gemini for routine code, and reviewed and reworked the output so the team could maintain it.

SKILLS

Frontend
React TypeScript Vue.js Nuxt.js Inertia.js HTML CSS
Tailwind CSS

Backend
Laravel Node.js (Express) FastAPI REST APIs
WebSockets (Reverb, Socket.io)

Data
MySQL PostgreSQL (pgvector) Redis

AI
Gemini API OpenAI Whisper RAG smolagents MCP

Tools
Git GitHub Actions Docker Compose PHPUnit Vitest Playwright
Figma

Basic
Java C++

LANGUAGES

Malay	Native	English	Fluent
Mandarin	Native		

PROJECTS

Web-Based Python Bootcamp **Final Year Project**
[github.com/Shuiweiqop/Web-Based-Python-Bootcamp-project](#)
Laravel, React (Inertia), MySQL, Gemini, Judge0

- Full-stack platform that teaches Python to beginners: placement test, lessons, coding exercises, quizzes, learning paths, a forum and gamification. 264 routes and 68 database migrations.
- Estimates how well each student knows each concept with a Bayesian Knowledge Tracing model, updated after every answer and weighted by question type and difficulty instead of a plain correct rate.
- Judge0 runs student Python code in a sandbox so it cannot touch the server; Gemini helps draft lessons and tag concepts.
- Wrote 338 automated tests (PHPUnit, Vitest, Playwright) that run on every push. They caught a double-counting bug that was corrupting the learning-gain figures.

Playground: Real-Time Multiplayer Games **Side Project**
[GitHub](#) | [Live demo](#)
Node.js (Express), Socket.io, PostgreSQL (Supabase), JWT

- Real-time multiplayer games (Draw & Guess, Werewolf). Each game is its own module, so adding one does not touch the others.
- Game state lives on the server. In Werewolf each player only receives what their role may see, so the browser cannot reveal other roles.
- JWT login plus a guest mode for joining without signing up; in-memory fallback so it runs locally without a database.

Meeting AI Platform **Side Project**
[github.com/Shuiweiqop/meeting-ai-platform](#)
Laravel, React (Inertia), MySQL, Redis, Laravel Reverb

- Upload a recording and get a transcript, summary and action items. Whisper does speech-to-text and Gemini writes the summary, in a background queue.
- Replaced polling with WebSocket push over Reverb, plus a timed check because Reverb can drop the connection silently.
- 5 MB chunked uploads with a progress bar to stop timeouts. MySQL and Redis in Docker Compose; 62 tests in CI on every push.

RAG Exam Q&A System **Side Project**
[github.com/Shuiweiqop/rag-exam-qa](#)
Python, FastAPI, React, PostgreSQL + pgvector, Gemini

- Measured retrieval with a 20-question eval set (Recall@1 85%, Recall@3 95%).
- Rebuilt retrieval as an agent tool (smolagents) so the model decides when to search, and exposed it as an MCP server.

CERTIFICATIONS

Hugging Face: AI Agents Fundamentals (2026)
Huawei HCIA: Datacom and Cloud Computing
Cisco CCNA: Introduction to Networks; Switching, Routing and Wireless Essentials; Enterprise Networking, Security and Automation